Comparing Unlike Fractions

Comparing with benchmarks, common denominators, and equivalent fractions

Directions:

- Two tools are practiced here: comparing each fraction to the benchmark $\frac{1}{2}$, and rewriting both fractions with a common denominator.
- When a blank asks for a symbol, write <, >, or =.

1. The strip below is cut into 8 equal parts, and half of it is shaded. Use the strip to fill in the missing numerator: $\frac{1}{2} = \frac{\square}{8}$

Answer: ____________________

2. Think about where $\frac{3}{8}$ sits on a number line from 0 to 1. Write the correct symbol: $\frac{3}{8}$ __________ $\frac{1}{2}$

3. Compare $\frac{2}{3}$ and $\frac{3}{4}$ by rewriting both in twelfths.
$\frac{2}{3} = \frac{\square}{12}$: __________ $\frac{3}{4} = \frac{\square}{12}$: __________ so $\frac{2}{3}$ __________ $\frac{3}{4}$

4. Lena's ribbon is $\frac{2}{5}$ m long. Her ruler is marked in tenths of a meter. Write a fraction with denominator 10 that names the same length.

Answer: ________________

5. Two juice jugs are the same size. Jug A is $\frac{5}{8}$ full and Jug B is $\frac{3}{8}$ full. Compare each jug to the benchmark $\frac{1}{2}$ to see which one is more than half full.

Jug A: $\frac{5}{8}$ ________ $\frac{1}{2}$ Jug B: $\frac{3}{8}$ ________ $\frac{1}{2}$

6. Pedro ran $\frac{5}{6}$ mi and Amara ran $\frac{7}{9}$ mi. Rewrite both distances in eighteenths of a mile to see who ran farther.

$\frac{5}{6} = \frac{\square}{18}$: ________ $\frac{7}{9} = \frac{\square}{18}$: ________ so $\frac{5}{6}$ ________ $\frac{7}{9}$

7. Compare $\frac{4}{9}$ and $\frac{6}{11}$ without finding a common denominator: $\frac{4}{9}$ ________ $\frac{6}{11}$.
Then explain how comparing each fraction to $\frac{1}{2}$ tells you which is greater.

8. Marco shades $\frac{2}{3}$ of one strip and $\frac{8}{12}$ of an identical strip.

(a) Fill in: $\frac{2}{3} = \frac{\square}{12}$: __________

(b) Write the correct symbol: $\frac{2}{3}$ __________ $\frac{8}{12}$

9. A pizza pan of brownies: Jon has $\frac{7}{12}$ of a pan left and Kayla has $\frac{5}{8}$ of a pan left. Rewrite both amounts in twenty-fourths to compare them.

$\frac{7}{12} = \frac{\square}{24}$: __________ $\frac{5}{8} = \frac{\square}{24}$: __________ so $\frac{7}{12}$ __________ $\frac{5}{8}$

10. Challenge.

(a) Write the correct symbol: $\frac{3}{8}$ __________ $\frac{1}{2}$

(b) Find a fraction that is greater than $\frac{3}{8}$ but less than $\frac{1}{2}$, and explain how you know it fits between them. __________